

CSA0614

DESIGN AND ANALYSIS OF ALGORITHMS

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Experiment 3: Factorial

Code:

```
def factorial_iterative(n):  
    fact = 1  
    for i in range(1, n + 1):  
        fact *= i  
    return fact  
  
def factorial_recursive(n):  
    if n == 0 or n == 1:  
        return 1  
    return n * factorial_recursive(n - 1)  
  
n = 5  
  
print("Iterative Factorial:", factorial_iterative(n))  
print("Recursive Factorial:", factorial_recursive(n))
```

Output:

```
===== RESTART: C:/Users,  
Iterative Factorial: 120  
Recursive Factorial: 120
```